EE2001 Electrical Circuits- I:

Continuation of the Topic:
Sinosoidal Steady-State Analysis

26 December 2023

Useful Supplementary Information

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- Useful supplementary information about fundamental operations with Complex Numbers and Phasors are available in 8 links provided below:

Link 1: https://www.electronics-tutorials.ws/accircuits/complex-numbers.html
Link 2: https://www.electronics-tutorials.ws/accircuits/phasors.html
Link 3: https://www.electronics-tutorials.ws/accircuits/ac-resistance.html
Link 4: https://www.electronics-tutorials.ws/accircuits/ac-inductance.html
Link 5: https://www.electronics-tutorials.ws/accircuits/phase-difference.html
Link 6: https://www.electronics-tutorials.ws/accircuits/ac-capacitance.html
Link 7: https://www.electronics-tutorials.ws/accircuits/sinusoidal-waveform.html
Link 8: https://www.electronics-tutorials.ws/accircuits/ac-waveform.html

The Sinusoidal Source

1

- We can translate a sine function to the cosine by subtracting 90° ($\pi/2$ rad) from the argument of the sine function:

$$\sin(\omega t + \theta) = \cos(\omega t + \theta - 90°)$$

Derivation:

$$\cos(\alpha - \beta) = cos\alpha\ cos\beta + sin\alpha\ sin\beta$$

Let $\alpha = \omega t + \theta$ and $\beta = 90°$

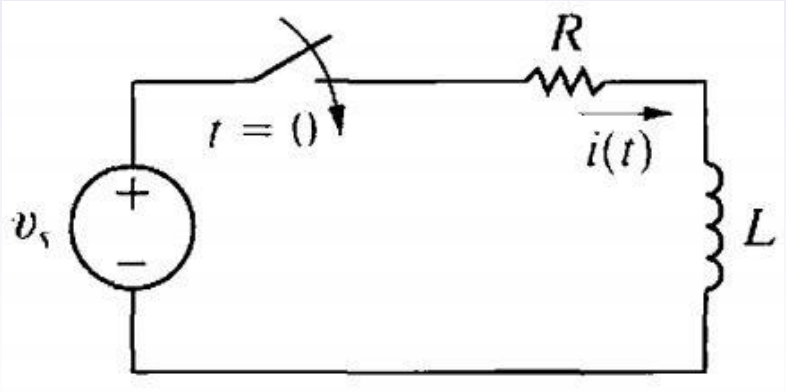
As $cos90° = 0$ and $\sin 90° = 1$, we have

$$\cos(\alpha - \beta) = sin\alpha = \sin(\omega t + \theta) = \cos(\omega t + \theta - 90°)$$

Example: $sin(\omega t + 30°) = \cos(\omega t + 30° - 90°) = \cos(\omega t - 60°)$

The Sinusoidal Response

2



- Before focusing on the steady-state response to sinusoidal sources, let's consider the problem in broader terms, in terms of the total response. This will help us to understand the steady-state solution perspective. v_s is a sinusoidal voltage:

$$v_s = V_m cos(\omega t + \phi)$$

- We assume the initial current in the circuit to be zero and measure time from the moment the switch is closed. The task is to derive the expression $i(t)$ for $t \geq 0$.
- It is similar to finding the step response of an RL circuit as in the previous lessons.
- The only difference is that the voltage source is now a time-varying sinusoidal voltage rather than a constant DC voltage. Application of KVL to the given circuit leads to the equation:

$$L\frac{di}{dt} + Ri = V_m(\omega t + \phi)$$

The Sinusoidal Response

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$$L\frac{di(t)}{dt} + Ri(t) = V_m(\omega t + \phi)$$

Can be solved for i as

$$i(t) = \frac{-V_m}{\sqrt{R^2 + \omega^2 L^2}}\cos(\varphi - \theta)\, e^{-(R/L)t} + \frac{V_m}{\sqrt{R^2 + \omega^2 L^2}}\cos(\omega t + \varphi - \theta)$$

- Where θ is the angle whose tangent is $\omega L/R$. Thus it can be easily calculated for a circuit driven by a sinusoidal source of known frequency.
- The first term on the right-hand side is referred to as the transient component of the current and it goes to infinity as time elapses.
- The second term on the right-hand side is known as the steady-state component of the solution. It exists as long as the switch remains closed and the source continues to supply the sinusoidal voltage.
- In this topic we develop a technique for calculating the steady-state response directly to avoid dealing with differential equation.

The Sinusoidal Response

2

- It is important to remember the following characteristics of the steady-state solution:
  - The steady-state solution is a sinusoidal function.
  - The frequnecy of the response signal is identical to the frequency of the source signal. This condition is always true in a linear circuit when the circuit parameters R, L and C are constant. Otherwise there is a nonlinear element in the circuit.
  - The maximum amplitude of the steady-state response, in general, differs from the maximum amplitude of the source. For the circuit given in the previous slides, the maximum amplitude of the response signal is $V_m/\sqrt{R^2+\omega^2L^2}$ and the maximum amplitude of the signal source is V_m
  - The phase angle of the response signal, in general, differs from the phase angle of the source. For the circuit given in the previous slides, the phase angle of the current is $\varphi-\theta$ and that of the voltage source is φ.
- When only finding the steady-state response, the task is reduced to finding the maximum amplitude and phase angle of the response signal. The waveform and frequency of the response are already known.

The Phasor

- The phasor is a complex number that carries the amplitude and the angle information of a sinusoidal function. It is based on Euler's identity, which related the exponential function to the trigonometric function

$$e^{\pm j\theta} = cos\theta \pm j\, sin\theta$$

- This is another way of expressing cosine and sine functions.
- We can consider the cosine function as the real part of the exponential function and the sine function as the imaginary part of the exponential function:

$$cos\theta = \mathrm{Re}\{e^{j\theta}\}$$
$$sin\theta = \mathrm{Im}\{e^{j\theta}\}$$

- Applying this principle

$$v(t) = V_m \cos(\omega t + \phi) = V_m Re\{e^{j(\omega t+\phi)}\} = V_m Re\{e^{j\omega t}e^{j\phi}\} = Re\{V_m e^{j\phi}e^{j\omega t}\}$$

- $V_m e^{j\phi}$ is a complex number that carries the amplitude and phase angle of the given sinusoidal function. This is the phase representation, or phasor transform of the given sinusoidal function.

The Phasor

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$$V = V_m e^{j\phi}\ (polar\ form) = V_m cos\phi + jV_m sin\phi\ (rectangular\ form)$$

- Both polar and rectangular forms are useful in circuit applications of the phasor concept.
- The frequent use of the exponential function $e^{j\phi}$ has led to an abbreviation called as angle notation. We extensively use the folowing notation:

$$e^{j\phi} = 1\angle\phi$$

- Inverse phasor transform example:

$$V = 100\angle -26° = 100\text{cost}(\omega t - 26°)$$

- The phasor transform is useful in circuit analysis, because it reduces the task of finding the maximum amplitude and phase angle of steady-state sinusoidal response to the algebra of complex numbers.
- The transient component vanishes as time elapses.
- Current can be expressed as $I_m e^{j\beta} = \frac{V_m e^{j\varphi}}{R + j\omega L}$

End of the Lesson (26.12.23)

Any questions?

New supplementary videos are added to the playlist on Youtube:

Supplementary videos: Enter Youtube-> Search for ‘Alparslan Zehir’ -> Go to Playlists -> Watch ‘Electrical Circuits’ playlist